INFORMATION GATHERING

PASSIVE RECONNAISSANCE

> Goal: Collect info without directly touching the target.
>
> **Focus:** Open-source intelligence (OSINT)

Domain & DNS Info

```
# WHOIS Lookup
whois target.com

# DNS Records (A, MX, TXT, NS)
dig target.com
dig -t MX target.com
dig -t TXT target.com
nslookup target.com
```

Subdomain Discovery

```
# Using crt.sh (Certificate Transparency)
curl -s "https://crt.sh/?q=%.target.com&output=json" | jq '.[].name_value'

# Using Sublist3r
sublist3r -d target.com

# Using amass
amass enum -d target.com
```

Google Dorking (OSINT)

```
site:target.com filetype:pdf confidential
site:target.com intitle:"index of"
site:target.com "password"
```

(Do this from browser or use `theHarvester`)

Company & Tech Stack Info

```
# Using theHarvester (emails, domains, subdomains)
theHarvester -d target.com -b all

# Using whatweb to identify technologies (passive)
whatweb target.com
```

Public Exposure Check

```
# Check leaks and breach data
haveibeenpwned.com
breachdirectory.org

# Check for public GitHub data
site:github.com "target.com" password
```

Shodan / Censys

```
# On shodan.io (web)
"target.com" org:"Target Company"
```

(or CLI if you have API key)

```
shodan search "target.com"
```

ACTIVE RECONNAISSANCE

> Goal: Interact directly with the target (detectable).
>
> **Focus:** Network scanning, service discovery, and probing.

Host Discovery

```
# Ping Sweep (discover live hosts)
nmap -sn 192.168.1.0/24
```

Port Scanning

```
# SYN Scan (most common)
nmap -sS -p 1-1000 target.com
```

```
# Version Detection
nmap -sV target.com

# OS Detection
nmap -O target.com

# All-in-one (Aggressive Scan)
nmap -A target.com
```

Service Enumeration

```
# Banner Grabbing (HTTP)
nc -v target.com 80
HEAD / HTTP/1.0

# SSH Banner
nc -v target.com 22
```

Web Application Scan

```
# Basic Vulnerability Scan
nikto -h http://target.com

# Directory Bruteforce
gobuster dir -u http://target.com -w /usr/share/wordlists/dirb/common.txt

# Using OWASP ZAP (GUI) or Burp Suite
```

SMB / FTP / SNMP Enumeration

```
# SMB Shares
smbclient -L \\target.com -N

# FTP Anonymous Login
ftp target.com
# (then try username: anonymous / password: anonymous)

# SNMP Enumeration
snmpwalk -v2c -c public target.com
```

Social Engineering (Human Recon)

- Call pretending to be vendor or support (vishing)
- Send phishing email (phishing simulation)